

Software Modeling and UML

Experiment 4

Activity Diagram Modeling

Objectives

- Understand the role of the activity diagrams
- Be familiar with the general process of modeling activity diagrams
- Become more proficient in using UML tools and master the representation of common modeling elements in activity diagrams

Case one: Activities in Library management system (Personal work)

- Construct the corresponding activity diagrams for the following cases:

- 1.Book Registration

- Upon the arrival of new books, the librarian will log into the system and check if the book information already exists. If not, they will register and record the books.

- 2.Borrow books

- The librarian will scan the campus card, check for overdue books, unpaid fines, and if the number of borrowing books exceeds the maximum limit. After confirming the above information, they will scan the barcode to complete the loan operation.

- You can define the activities in as much detail as you find reasonable.

Case two: Human resource management system (Personal work)

- Construct the corresponding activity diagrams for the following cases:
 - 1. Recruitment related activities
 - Recruitment refers to the process of identifying, attracting, interviewing, selecting, hiring and onboarding employees.
 - 2. Employee leave management
 - Leave management is the process of approving, tracking, and controlling leave applications through effective management. It helps HR in retaining employee satisfaction while ensuring the availability of enough workforce at all times.
- You can define the activities in as much detail as you find reasonable.

Case three: Activities in Research project management system (Personal work)

- Construct the corresponding activity diagrams for the following cases
 - 1. User registration
 - Project applicants need to register through the system. Various departments within the company need to verify the applicant's identity, credit, research, and other aspects. The registration is considered complete only when all the aforementioned audits are approved.
 - 2. Project audit
 - Project applicants are required to submit their applications through the system. Once the project passes the initial assessment, it proceeds to the expert review stage. Following expert evaluation, if the project is approved, it moves forward for implementation and undergoes mid-assessment and project conclusion stages.
- You can define the activities in as much detail as you find reasonable.

Please consider the following questions

- In which view can an activity diagram be created?
- What is the purpose of an activity diagram?
- What is the purpose of a swimlane, and how is it used?
- How can concurrent activities be expressed?
- How can the details of complex activities be represented within an activity node, and how can a sub-activity diagram be created?
- Is only one activity diagram used when building a software model?

Steps of Activity Diagram Modeling

- Define the scope of the activity diagram
- Add the initial node and final node
- Add activity nodes
- Add decision nodes
- Identify parallel activities
- Add control flows

Experiment requirements

- Please send your work through the link provided by the teaching assistant's cloud storage.
- Requirements for the report
 - Report name: Student ID Name UML EX4 Activity diagram case name
 - Please do not compress any document (不要压缩)
 - The deadline for task submission is 6 PM this Saturday evening
- Make sure to send the report following the specified guidelines; otherwise, we will not be able to receive your submission.